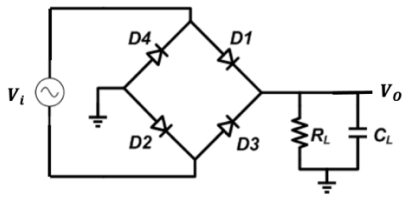
1) The circuit below uses diodes with $V_{DO} = 0.8\,V$. The load resistor and capacitor are 1 $k\Omega$ and 47 $\mu$F, respectively. The input signal is given by $V_i = V_S \sin 2\pi 60t$




(a) Calculate the expected ripple voltage of the circuit.

$$V_r = \frac{V_s - 2V_{DO}}{2fRC} = \frac{V_s - 2(0.8)}{2(60)(1000)(47 \times 10^{-6})}$$

(b) Calculate the average output voltage.

$$v_{o_ave} = V_s - 2(0.8) - \frac{V_r}{2}$$

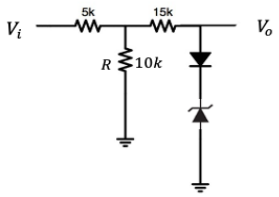
(c) What would be the ripple value and the average output voltage if $D_2$ was burned?

The rectifier will work as a half-wave rectifier but with two diodes in series.

$$V_r = \frac{V_s - 2V_{DO}}{fRC} = \frac{V_s - 2(0.8)}{(60)(1 \times 10^3)(47 \times 10^{-6})}$$

$$v_{o_ave} = V_s - 2(0.8) - \frac{V_r}{2}$$

2) For the diode circuit shown below, the Zener diode has $V_{ZO} = 3\,V$ and $r_z = 0\,\Omega$. Take $V_{DO} = 0.7\,V$ for both diodes. Show all steps and calculations. $V_i = V_S \sin 2\pi 60t$.

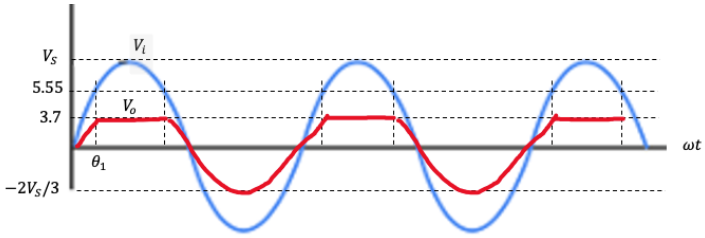



(a) Draw the output waveform ($V_O$ vs. $t$). Show all labels on the curve.

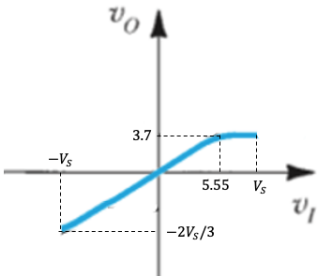
The diodes will need (0.7+3) V to switch ON. To provide this voltage, the input should reach to $V_i = \left(\frac{15}{10}\right)(3.7) = 5.55\,V$.

For $V_i \geq 5.55\,V$, the output is fixed at 3.7 V. Otherwise, the diodes are OFF and the output is $V_o = \left(\frac{10}{15}\right) V_i$

$$\theta_1 = \sin^{-1}\left(\frac{5.55}{V_s}\right)$$



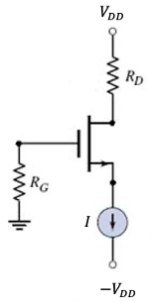

(b) Draw the transfer function ($V_O$ vs. $V_i$) of the circuit. Show all labels on the curve.




(c) Find the maximum current magnitude through the resistor $R$.

The maximum current happens when the diodes are OFF. $I_{R_max} = V_s\left(\frac{10}{15}\right)/10 = \frac{V_s}{15\,k}$

3) The MOSFET in the circuit below has $V_t = 1$ V, $k$=1 mA/V$^2$ and λ=0. Given $I$=2 mA, find the values of $R_D$ and $R_G$ that will put the transistor at the edge of saturation mode.




Edge of saturation is $V_{DS} = V_{GS} - V_t \quad \rightarrow \quad V_D = -V_t = -1$ V, because $V_G = 0$.

$I_D = 2\ mA \quad \rightarrow R_D = \frac{V_{DD} - V_D}{I_D} = \frac{V_{DD}+1}{2\ m}$ $\qquad R_G$ can be any value, because the gate current is zero.

In your design, find the voltage at the transistor source terminal. Show all steps and calculations.

$$I_D = k(V_{GS} - V_t)^2$$

$$2\ m = 1\ m(-V_S - 1)^2$$

$$-V_S - 1 = \pm\sqrt{2}$$

$$V_S = -1 - \sqrt{2} = -2.41 \text{ V}.$$

(The other solution corresponds to cutoff mode)